Promo Finder: finding 20-60s promo clips in screenplays

Kuku AI Generalist Assignment, Problem 1. Tested on "I Married a Broken Soldier. He Was a Duke." (50 episodes)

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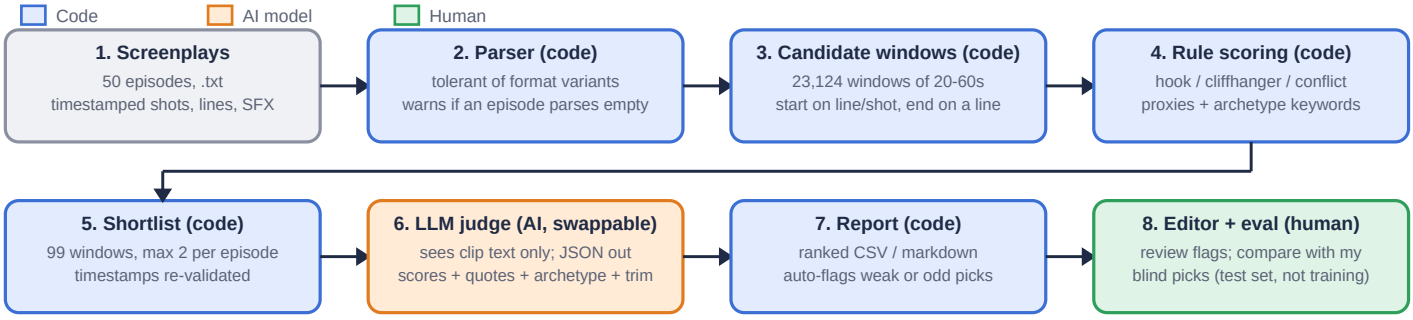
Why Problem 1, and what I assumed. It is the only problem with real input and an output I can check: I read five episodes myself, hand-picked 12 promo segments *before* seeing any tool output, and used them as a test set. The brief was silent on several points, so I assumed: (1) a promo is one continuous 20-60s window that opens on a line or shot and ends on a spoken line (cross-episode stitching is out of scope); (2) the screenplay text is all the tool sees: no video, acting or music; (3) timestamps are seconds inside the episode; (4) at most two promos per episode is a sensible shortlist; (5) there are no labels in the pack, so nothing is trained: the tool is rules plus a zero-shot LLM judge, and my picks only test it.

How the pipeline works



Score (0-100) = hook 25 + cliffhanger 30 + standalone clarity 20 (LLM) + conflict 15 + pacing 10 (rules), each part 0-10, shown in full for every promo. Every promo also carries an **archetype** (Public Humiliation / Betrayal, Hidden Identity Reveal, Revenge / Comeuppance, Forbidden Romance, Danger / Life-or-Death, Secret / Conspiracy, or Power Showdown as fallback) so an editor can read its type at a glance.

What AI, code and a human each do

Who	Job	Guard against failure
Code	Parse timestamps; build 23,124 candidate windows; cheap rule scores; rank; cache; report.	Parser skips non-episode files and exits if none parse. Every window ends on a real line boundary. Timestamps re-checked against the source.
AI (Gemini; OpenRouter or mock via config.json)	Reads *only the clip text* (no logline, no earlier episodes) like a cold viewer. Returns JSON: summary, who is who, hook / cliffhanger / standalone scores, archetype, best start and end line.	The model picks *line numbers*, never timestamps; code converts them. Evidence quotes must appear verbatim in the clip or the answer is rejected and retried. Archetype must be from a fixed list. Trim re-checked to 20-60s.
Human	Editor reviews auto-flagged clips (weak hook or cliffhanger, low standalone, odd trim, archetype disagreement). I wrote the test set blind.	Approve/reject decisions would become a growing labelled set.

Results (honest numbers)

My 12 hand-picked segments, 5 episodes	Rules only (99-clip pool)	With LLM judge (10-clip pool)
Segment at least 80% inside a tool window	7 / 12	7 / 12
Tight overlap (IoU >= 0.5) / loose (IoU >= 0.3)	3 / 12 and 8 / 12	3 / 12 and 9 / 12
Tool rank of my top-3 picks (#1 Ep22, #2 Ep33, #3 Ep47)	17, 66, 23 of 99	2, 6, 1 of 10. Rules rank the *same* 10 clips 1, 7, 2

Read this carefully. The LLM was only run on 15 of 99 shortlisted clips (free-tier quota, see page 2), so the two columns use different pools. Inside the same 10 clips, the LLM ordered my top three almost exactly as the rules did, so I cannot claim the LLM improved ranking against my picks. What it did change: it reordered the rules top 5 (Ep38, a prison monologue I judged weak by hand, fell from 82.0 to 62.5; Ep32, a hot iron stopping millimeters from the heroine eye, rose to #1 at 81.2). 12 picks over 5 episodes is a rough signal, not a benchmark, and I ran everything once, so run-to-run stability is untested.

Where the tool picks badly (real examples)

Failure	Evidence from the runs
Misses opening hooks	No candidate near my Ep22 19-42s or Ep33 2-29s picks (no tool window overlaps them, before or after the LLM). Cause: max 2 windows per episode and no rule for episode openings. **Not fixed.**
Rules read punctuation, not menace	Cliffhanger rule scored 0 for 11 of my 12 windows because they end on plain sentences ("You will scrub the floors you walked on as an impostor"), not on ? or !. That is why rules ranked my picks 17, 66 and 23 of 99.
LLM inflates thin hooks / resolved beats	Ep5: hook "Miss Ashcroft." scored 9/10. Ep47: "I accept, Your Grace." got cliffhanger 7 with an empty unresolved-question field (a resolved beat). Ep14: cliffhanger "evidence" is a camera description, not a line.
Two scenes glued together	Ep47 clip (23.4-66.8s) switches from Juliana and Alexander to Lady Ashbourne and Catherine. The LLM flagged it (standalone 5/10); the rules could not see it. Needs a scene-boundary rule.
Archetype labels disagree	Keyword rules and the LLM agree on only 2 of 15 clips. I trust the LLM label more but have not checked all 15 by hand. Under the rules, 38 of 99 clips fell into the fallback.
Run limits	Only 15 of 99 clips judged: 5 by gemini-3.5-flash, 10 by gemini-flash-lite-latest, so their scores are not strictly comparable. No second-model comparison. Text only: acting, visuals and music are invisible.

Getting Gemini working: what broke and how I fixed it

What happened	Cause	Fix
Episode 6 parsed as zero scenes	Headers in ALL CAPS in one file	Case-insensitive headers; check that every episode yields scenes
12 of 50 episodes missing from the shortlist	I had capped output at 40 promos; I noticed when asking why only 40	Cap removed: 99 promos cover 50 of 50 episodes
HTTP 404, model not found	AI assistant hardcoded gemini-2.5-flash from memory; retired for new accounts	Model name moved to config.json
90s timeouts on gemini-3.8-flash	Name taken from an error message; not in my key model list	test_gemini.py lists models the key can use; switched to a listed model
HTTP 429 at clip 20 of 99	Burst calls hit the free-tier quota	7s pacing, 45s backoff, stop after 3 in a row, --episodes to judge priority clips first
Wrong folder, .env lost its dot, key pasted in chat	Setup mistakes	Key rotated; key lives only in .env (git-ignored); clear "run X first" errors

Scale and cost

Per show the LLM stage is about 99 calls of roughly 1.3k tokens in and 0.4k out, around 170k tokens: on a Flash-class model that is small money (check current pricing); in my run the free quota stopped me after about 20 clips on one model, so a paid or Kuku-provided key is needed at one new show per week. Stage 1 costs nothing and runs in seconds, cutting 23,124 windows to 99 before any model call. Calls are cached by model plus prompt and resumable. To scale: judge only each show top 30, add an opening-window rule and a scene-boundary rule, add pairwise ranking so scores do not cluster, and use editor approve/reject as a growing eval set.

How I used AI, and where it got it wrong

Tools: Claude as pair-programmer (parser, scoring, judge, this draft), Gemini as the runtime judge, VS Code to run it. **My decisions:** chose Problem 1; wanted to "train on all 50 promos", then learned the pack has no labels, so I made my own blind picks the test set; asked for archetypes and a visible score breakdown; questioned whether case-folding would erase hook signal (it only touches headers; dialogue has no all-caps words, so punctuation is the signal I used); caught the 40-promo cap bug. **One place the AI got it wrong:** it hardcoded a Gemini model name from memory that had been retired, then, when I hit the 404, suggested another name straight from an error message that was not in my account model list, which timed out for 90s. I fixed it by asking the API which models my key could use, and I now keep the model in config, not in code.

Prototype: GitHub repo with run steps plus a Streamlit app (upload episodes, pick model, run, download). The confidential pack is not in the repo; the raw outputs folder goes with the submission form only.